Δ.72 Coherence Engine, Open-Data KPI Proxies

Expedite, Overtime, Logistics, quarterly mean signals, 2020-Q1 to 2025-Q3

KPI proxy (open data)	Base sMAPE	Δ.72 sMAPE	sMAPE ↓	QoQ var ↓	THD ↓	Shock qtr	Recovery (base→Δ.7
Logistics volume proxy (Freight TSI, FRED)	1.38%	1.08%	21.6%	89.9%	0.455	2020-Q2	NA→NA qtrs
Logistics shipments proxy (Cass Freight Shipments, FRED)	5.79%	4.78%	17.4%	85.8%	0.517	2020-Q2	NA→NA qtrs
Logistics cost proxy (PPI: Delivery & Warehousing, FRED)	7.06%	8.02%	-13.5%	86.8%	1.311	2023-Q2	1→0 qtrs
Overtime pressure proxy (Manufacturing OT hours, FRED)	5.43%	4.64%	14.6%	90.1%	0.291	2020-Q2	NA→NA qtrs
Expedite likelihood proxy (Delivery time diffusion, Philly Fed via FRED)	143.41%	147.59%	-2.9%	94.8%	0.613	2022-Q3	2→0 qtrs
Expedite cost proxy (PPI: Couriers & Express, FRED)	8.30%	9.45%	-13.9%	93.4%	0.976	2020-Q2	NA→NA qtrs

Interpretation

These are public time series that correlate with the same operational forces that drive expedite spend, overtime hours, and logistics variability. Because some series can cross zero, this report uses sMAPE (symmetric MAPE) for forecast error. Δ.72 is evaluated on sMAPE improvement, variance reduction on quarter-to-quarter deltas, and harmonic distortion reduction (THD proxy).

Charts

Logistics volume proxy (Freight TSI, FRED)

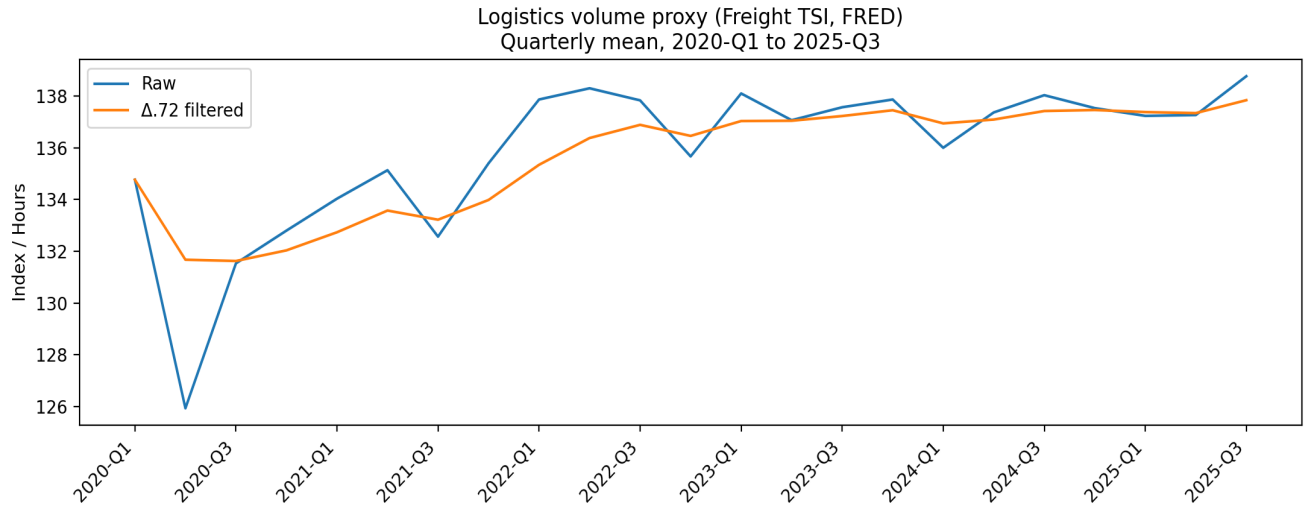


Figure: raw vs Δ.72 filtered (state estimate).

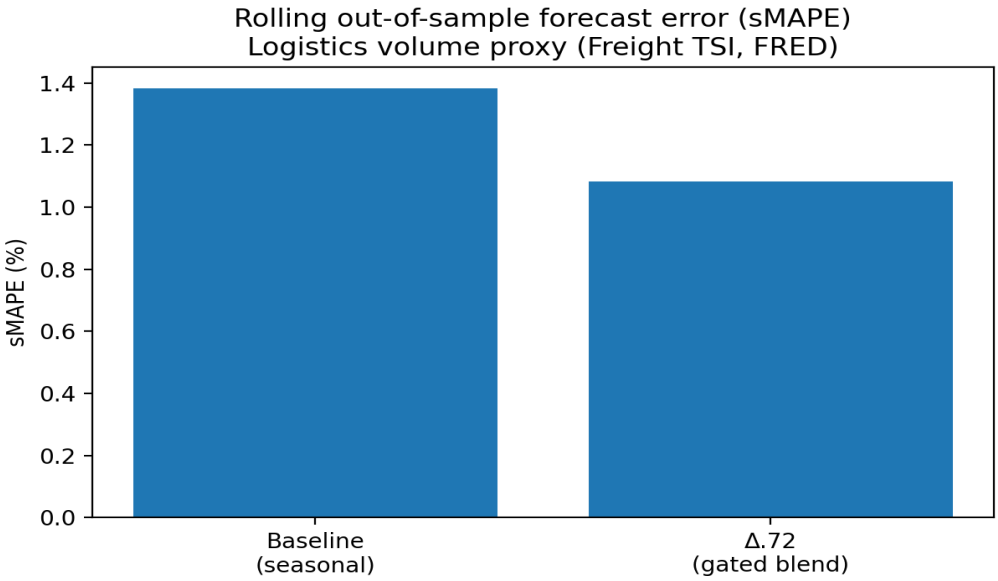
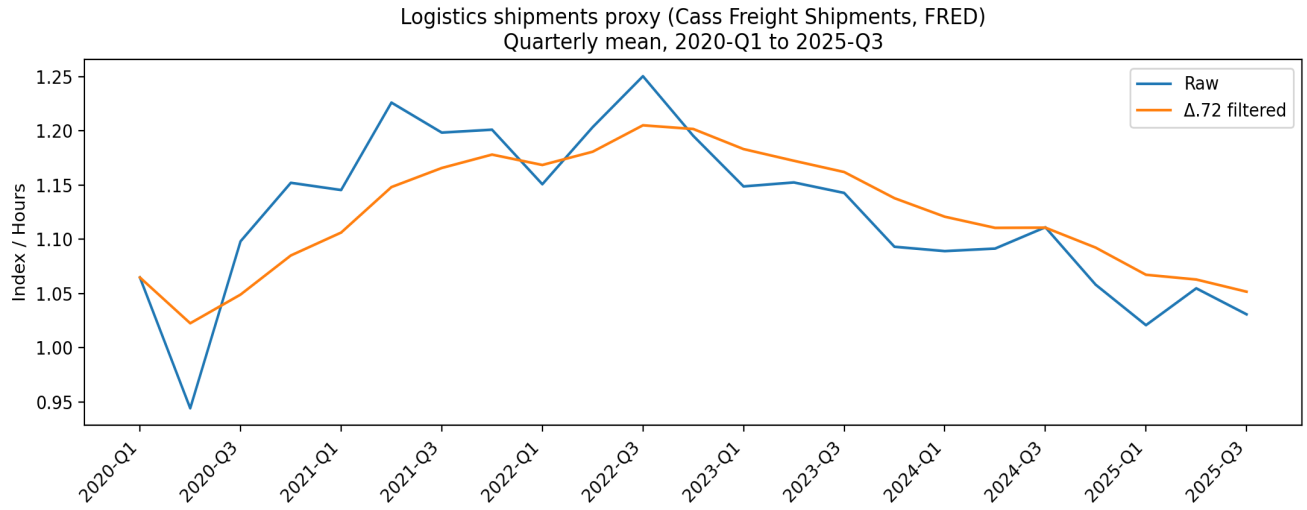


Figure: rolling out-of-sample forecast error (sMAPE), baseline seasonal vs Δ.72 gated blend.

Logistics shipments proxy (Cass Freight Shipments, FRED)



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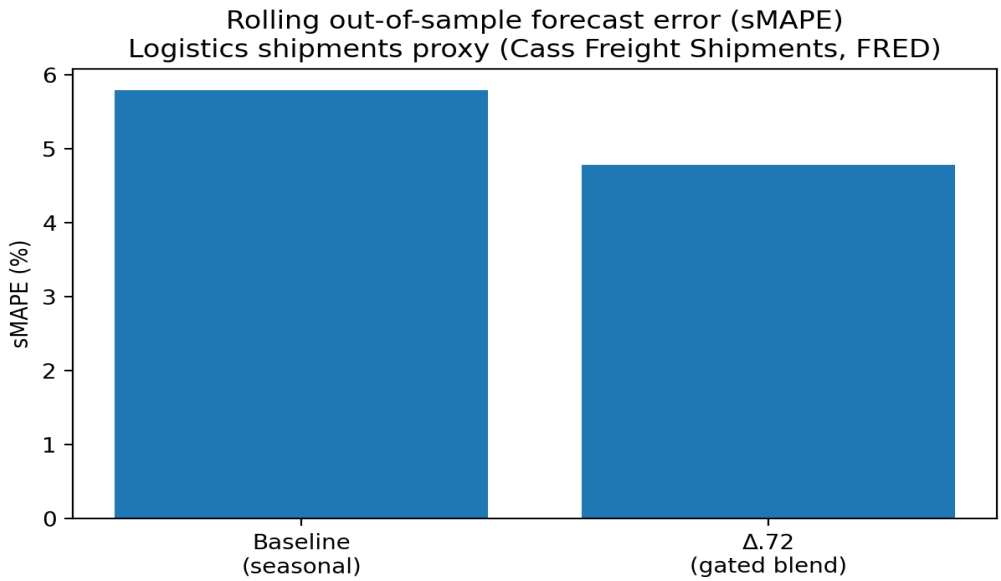
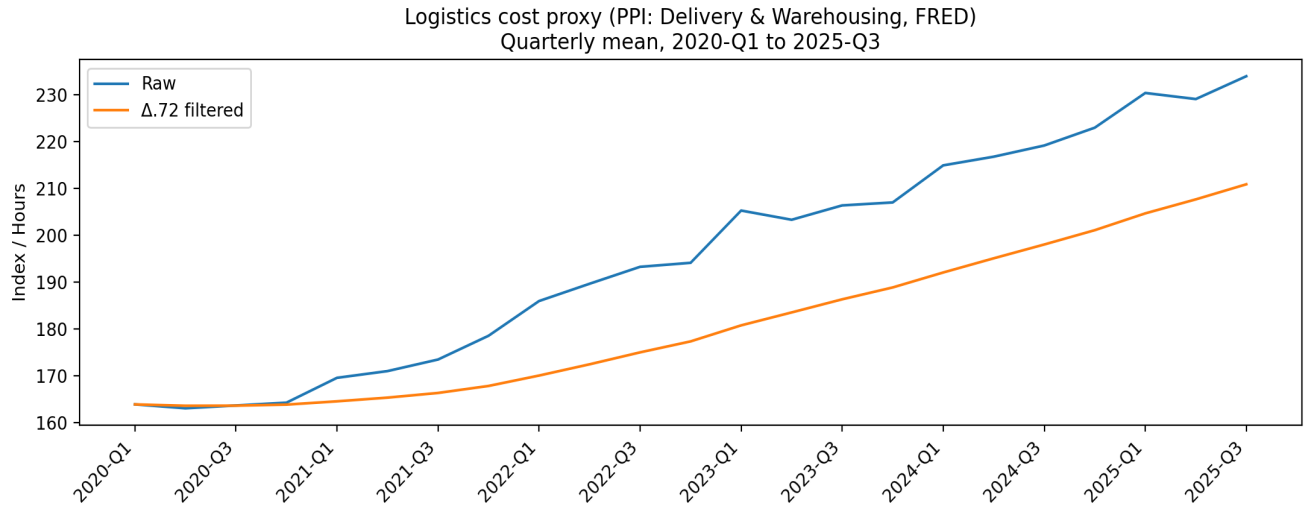


Figure: rolling out-of-sample forecast error (sMAPE), baseline seasonal vs Δ.72 gated blend.

Logistics cost proxy (PPI: Delivery & Warehousing, FRED)



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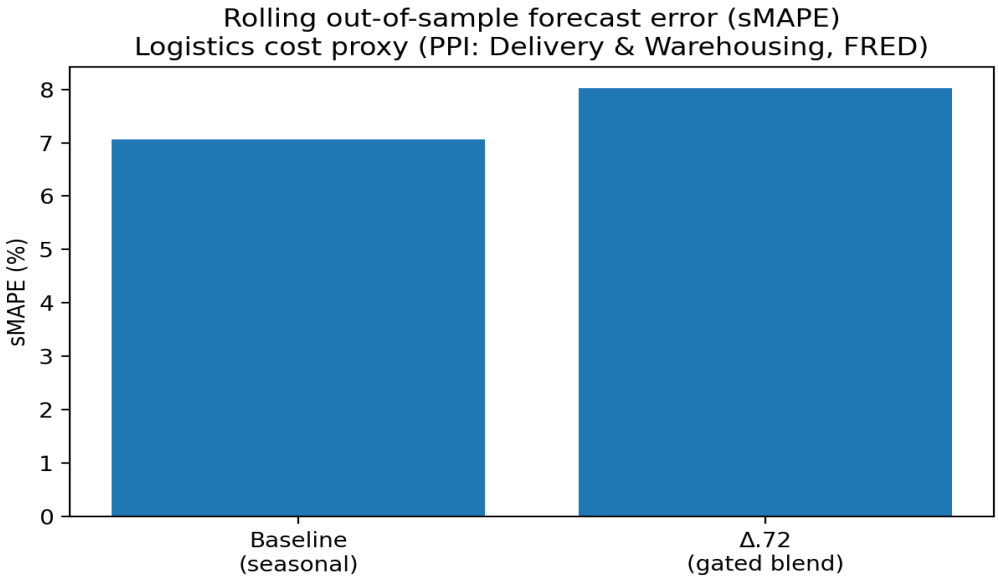


Figure: rolling out-of-sample forecast error (sMAPE), baseline seasonal vs Δ.72 gated blend.

Overtime pressure proxy (Manufacturing OT hours, FRED)

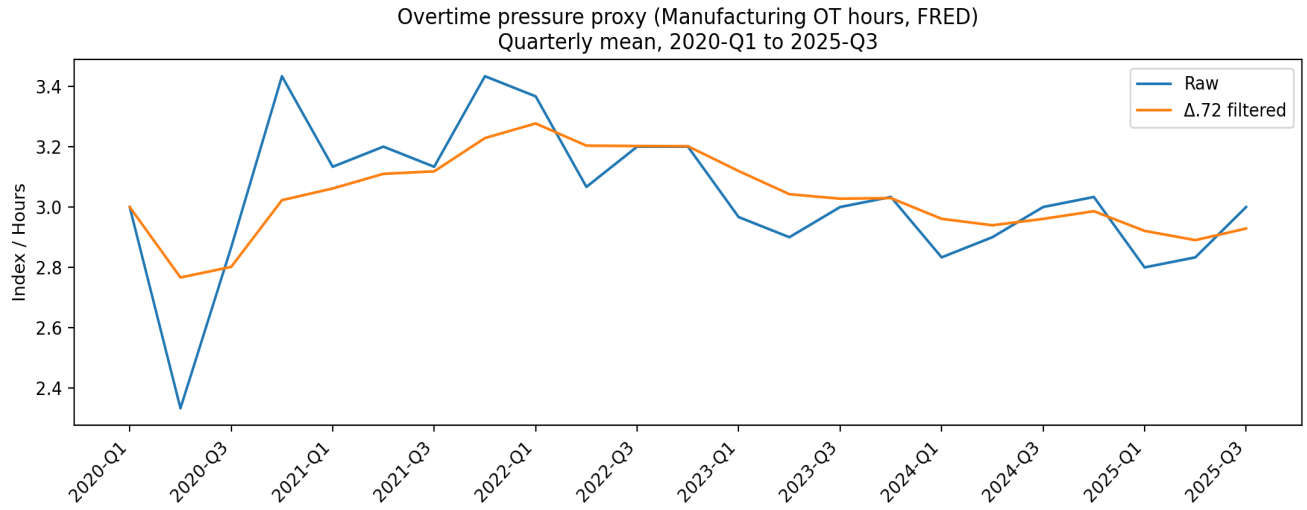


Figure: raw vs Δ.72 filtered (state estimate).

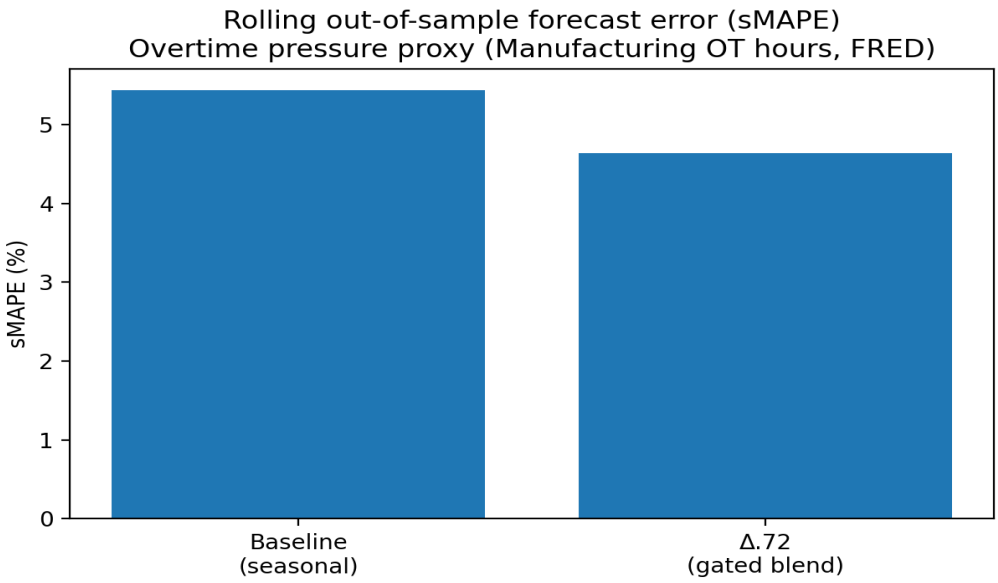


Figure: rolling out-of-sample forecast error (sMAPE), baseline seasonal vs Δ.72 gated blend.

Expedite likelihood proxy (Delivery time diffusion, Philly Fed via FRED)

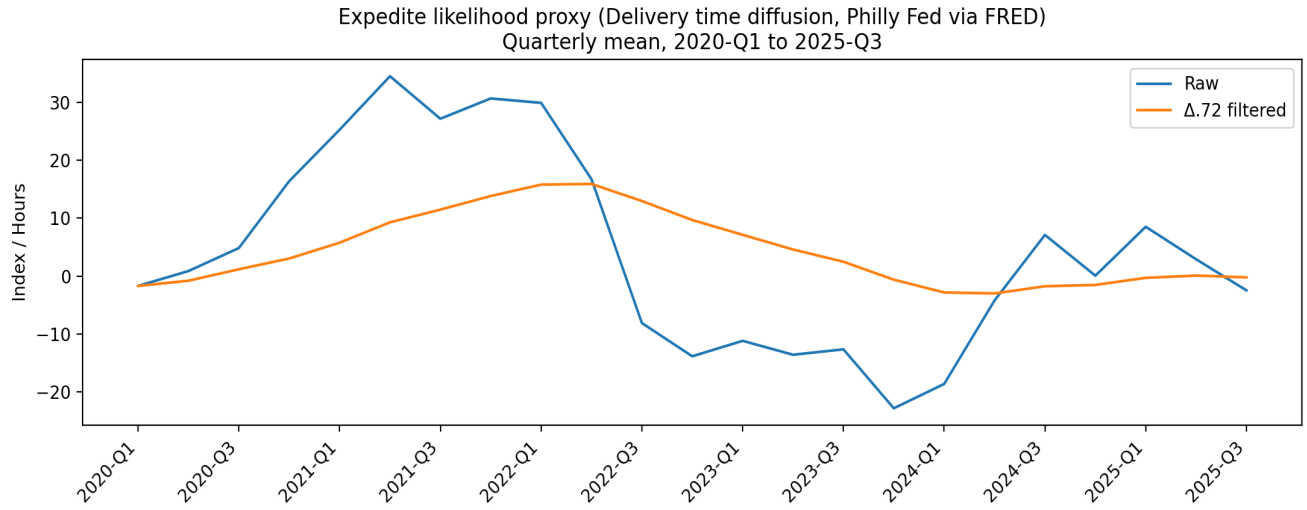
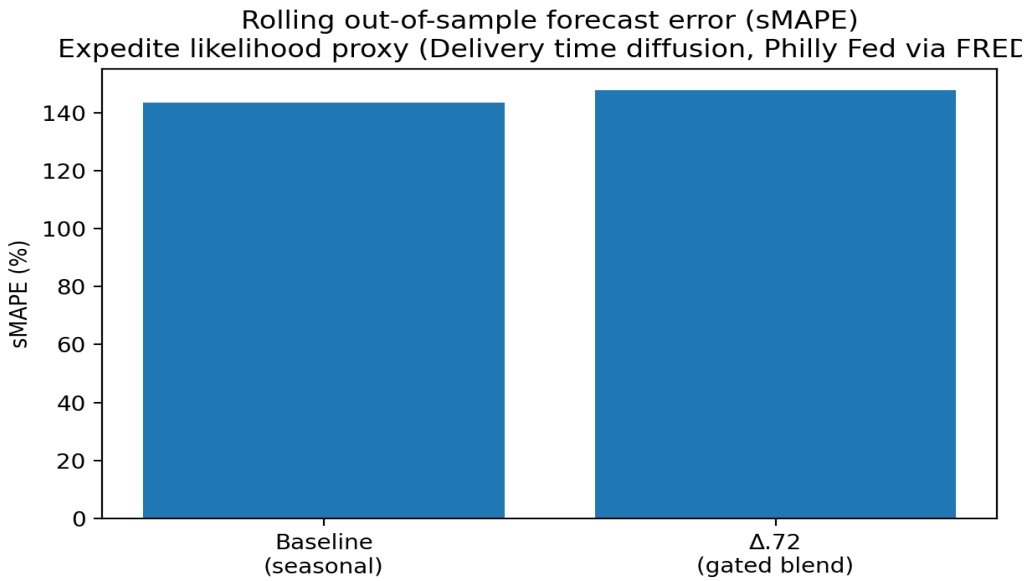


Figure: raw vs Δ.72 filtered (state estimate).



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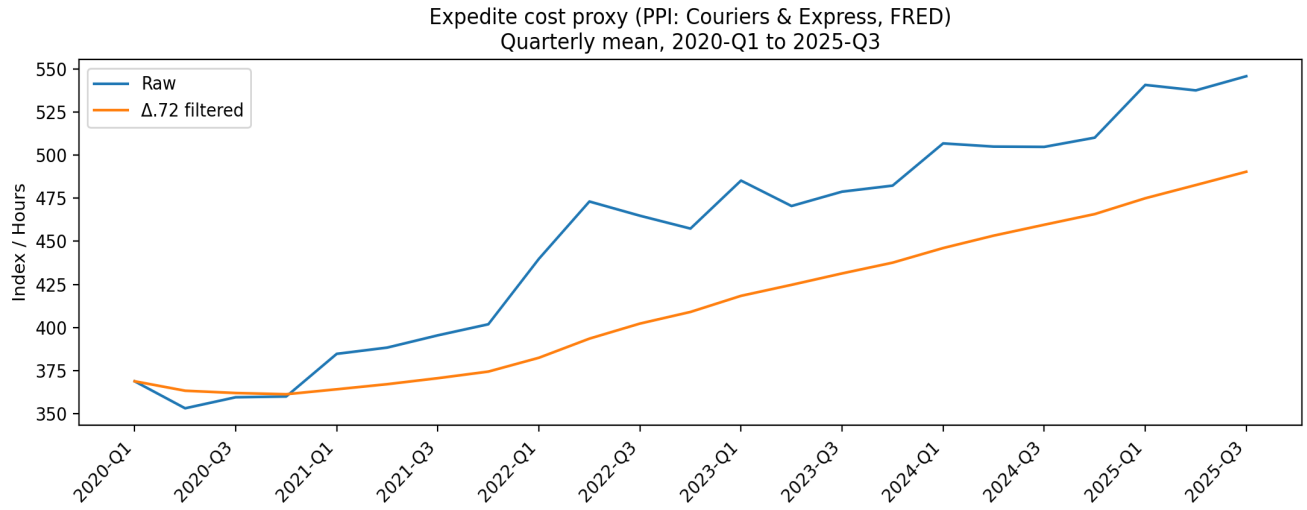
Expedite cost proxy (PPI: Couriers & Express, FRED)



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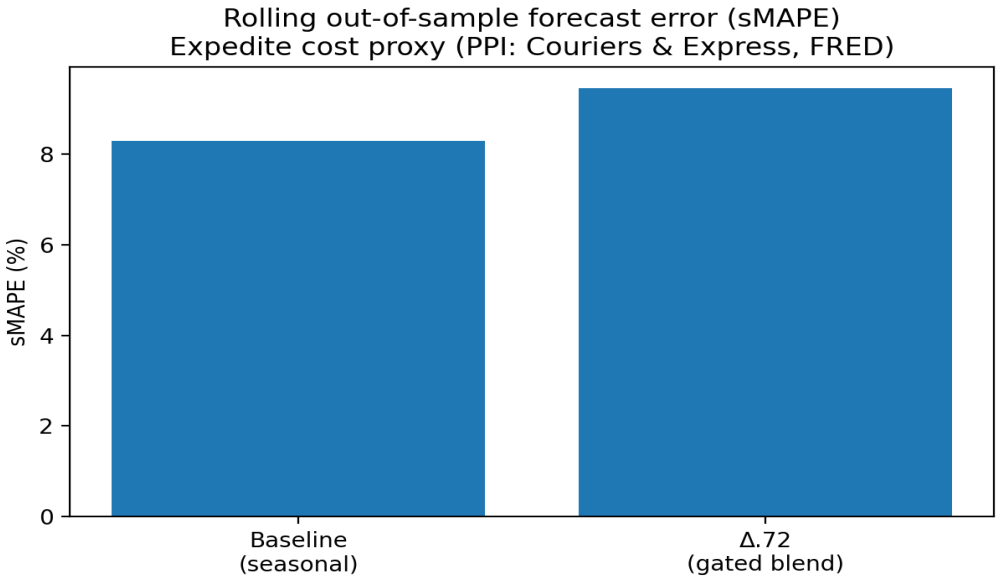

Figure: rolling out-of-sample forecast error (sMAPE), baseline seasonal vs Δ.72 gated blend.

Appendix, Backtest tables

Backtest detail, Logistics volume proxy (Freight TSI, FRED)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2020-Q1	134.8					
2020-Q2	125.9					
2020-Q3	131.5					
2020-Q4	132.8					
2021-Q1	134	134.8	133.5	0.55	0.37	
2021-Q2	135.1	125.9	129	7.05	4.65	
2021-Q3	132.6	131.5	132.5	0.78	0.09	
2021-Q4	135.4	132.8	133	1.94	1.80	
2022-Q1	137.9	134	134	2.82	2.84	
2022-Q2	138.3	135.1	135.2	2.32	2.25	
2022-Q3	137.8	132.6	134.3	3.90	2.61	
2022-Q4	135.7	135.4	136.1	0.20	0.30	
2023-Q1	138.1	137.9	137.2	0.17	0.63	
2023-Q2	137.1	138.3	137.7	0.90	0.48	
2023-Q3	137.6	137.8	137.5	0.19	0.06	
2023-Q4	137.9	135.7	136.4	1.61	1.09	
2024-Q1	136	138.1	137.8	1.53	1.32	
2024-Q2	137.4	137.1	137	0.22	0.26	

Backtest detail, Logistics volume proxy (Freight TSI, FRED) (continued)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2024-Q3	138	137.6	137.4	0.34	0.49	
2024-Q4	137.5	137.9	137.7	0.24	0.10	
2025-Q1	137.2	136	136.7	0.90	0.42	
2025-Q2	137.3	137.4	137.4	0.07	0.08	
2025-Q3	138.8	138	137.7	0.53	0.76	

Backtest detail, Logistics shipments proxy (Cass Freight Shipments, FRED)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2020-Q1	1.065					
2020-Q2	0.944					
2020-Q3	1.098					
2020-Q4	1.152					
2021-Q1	1.145	1.065	1.074	7.30	6.45	
2021-Q2	1.226	0.944	1.017	25.99	18.64	
2021-Q3	1.198	1.098	1.121	8.74	6.71	
2021-Q4	1.201	1.152	1.158	4.16	3.63	
2022-Q1	1.151	1.145	1.16	0.46	0.81	
2022-Q2	1.203	1.226	1.2	1.87	0.27	
2022-Q3	1.25	1.198	1.19	4.25	4.91	
2022-Q4	1.195	1.201	1.203	0.47	0.62	
2023-Q1	1.149	1.151	1.174	0.17	2.15	
2023-Q2	1.152	1.203	1.194	4.33	3.57	
2023-Q3	1.143	1.25	1.215	9.00	6.16	
2023-Q4	1.093	1.195	1.18	8.94	7.68	
2024-Q1	1.089	1.149	1.144	5.33	4.91	
2024-Q2	1.091	1.152	1.138	5.44	4.20	

Backtest detail, Logistics shipments proxy (Cass Freight Shipments, FRED) (continued)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2024-Q3	1.111	1.143	1.128	2.81	1.53	
2024-Q4	1.058	1.093	1.101	3.25	3.98	
2025-Q1	1.021	1.089	1.09	6.48	6.61	
2025-Q2	1.055	1.091	1.08	3.42	2.42	
2025-Q3	1.031	1.111	1.089	7.50	5.53	

Backtest detail, Logistics cost proxy (PPI: Delivery & Warehousing, FRED)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2020-Q1	163.9					
2020-Q2	163.1					
2020-Q3	163.7					
2020-Q4	164.3					
2021-Q1	169.6	163.9	163.9	3.40	3.41	
2021-Q2	171	163.1	163.5	4.75	4.48	T
2021-Q3	173.5	163.7	164.2	5.81	5.51	T
2021-Q4	178.5	164.3	164.9	8.33	7.95	T
2022-Q1	185.9	169.6	169	9.21	9.52	T
2022-Q2	189.6	171	170.7	10.33	10.50	T
2022-Q3	193.2	173.5	173.2	10.79	10.96	T
2022-Q4	194.1	178.5	177.5	8.35	8.95	T
2023-Q1	205.2	185.9	183.4	9.87	11.26	T
2023-Q2	203.3	189.6	187	6.95	8.36	T
2023-Q3	206.3	193.2	190.3	6.56	8.08	T
2023-Q4	207	194.1	191.8	6.42	7.63	T
2024-Q1	214.9	205.2	200.3	4.59	7.01	T
2024-Q2	216.7	203.3	199.9	6.41	8.08	T

Backtest detail, Logistics cost proxy (PPI: Delivery & Warehousing, FRED) (continued)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2024-Q3	219.1	206.3	203	6.01	7.66	T
2024-Q4	222.9	207	204.3	7.42	8.73	T
2025-Q1	230.3	214.9	210.7	6.94	8.88	T
2025-Q2	229	216.7	213.1	5.52	7.20	T
2025-Q3	233.9	219.1	215.7	6.53	8.11	T

Backtest detail, Overtime pressure proxy (Manufacturing OT hours, FRED)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2020-Q1	3					
2020-Q2	2.333					
2020-Q3	2.867					
2020-Q4	3.433					
2021-Q1	3.133	3	3.01	4.35	4.01	
2021-Q2	3.2	2.333	2.661	31.33	18.39	
2021-Q3	3.133	2.867	2.976	8.89	5.15	
2021-Q4	3.433	3.433	3.291	0.00	4.22	
2022-Q1	3.367	3.133	3.176	7.18	5.82	
2022-Q2	3.067	3.2	3.235	4.26	5.33	
2022-Q3	3.2	3.133	3.165	2.11	1.11	
2022-Q4	3.2	3.433	3.329	7.04	3.96	
2023-Q1	2.967	3.367	3.292	12.63	10.40	
2023-Q2	2.9	3.067	3.09	5.59	6.35	
2023-Q3	3	3.2	3.129	6.45	4.21	
2023-Q4	3.033	3.2	3.122	5.35	2.89	
2024-Q1	2.833	2.967	2.995	4.60	5.55	
2024-Q2	2.9	2.9	2.927	0.00	0.94	

Backtest detail, Overtime pressure proxy (Manufacturing OT hours, FRED) (continued)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2024-Q3	3	3	2.973	0.00	0.91	
2024-Q4	3.033	3.033	3.001	0.00	1.08	
2025-Q1	2.8	2.833	2.902	1.18	3.58	
2025-Q2	2.833	2.9	2.909	2.33	2.65	
2025-Q3	3	3	2.951	0.00	1.66	

Backtest detail, Expedite likelihood proxy (Delivery time diffusion, Philly Fed via FRED)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2020-Q1	-1.7					
2020-Q2	0.8667					
2020-Q3	4.833					
2020-Q4	16.37					
2021-Q1	25.23	-1.7	-0.2804	200.00	200.00	T
2021-Q2	34.47	0.8667	2.332	190.19	174.65	T
2021-Q3	27.13	4.833	6.164	139.52	125.95	T
2021-Q4	30.63	16.37	14.89	60.71	69.14	T
2022-Q1	29.87	25.23	21.81	16.82	31.20	T
2022-Q2	16.7	34.47	28.86	69.45	53.38	T
2022-Q3	-8.1	27.13	23.76	200.00	200.00	T
2022-Q4	-13.83	30.63	25.33	200.00	200.00	T
2023-Q1	-11.17	29.87	23.81	200.00	200.00	T
2023-Q2	-13.57	16.7	13.82	200.00	200.00	T
2023-Q3	-12.63	-8.1	-4.295	43.73	98.51	T
2023-Q4	-22.8	-13.83	-8.941	48.95	87.32	T
2024-Q1	-18.6	-11.17	-8.003	49.94	79.66	T
2024-Q2	-4.2	-13.57	-10.34	105.44	84.49	T

Backtest detail, Expedite likelihood proxy (Delivery time diffusion, Philly Fed via FRED) (continued)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2024-Q3	7.1	-12.63	-9.741	200.00	200.00	T
2024-Q4	0.06667	-22.8	-16.49	200.00	200.00	T
2025-Q1	8.5	-18.6	-13.48	200.00	200.00	T
2025-Q2	2.9	-4.2	-3.031	200.00	200.00	T
2025-Q3	-2.467	7.1	4.996	200.00	200.00	T

Backtest detail, Expedite cost proxy (PPI: Couriers & Express, FRED)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2020-Q1	368.8					
2020-Q2	353.2					
2020-Q3	359.6					
2020-Q4	360					
2021-Q1	384.7	368.8	365.5	4.22	5.14	
2021-Q2	388.4	353.2	356.5	9.49	8.56	T
2021-Q3	395.5	359.6	361.8	9.51	8.88	T
2021-Q4	401.9	360	363.2	11.00	10.12	T
2022-Q1	439.8	384.7	381.6	13.36	14.16	T
2022-Q2	473.1	388.4	386.6	19.66	20.12	T
2022-Q3	464.8	395.5	394.9	16.12	16.27	T
2022-Q4	457.4	401.9	402	12.91	12.88	T
2023-Q1	485.2	439.8	430.6	9.81	11.93	T
2023-Q2	470.5	473.1	456.7	0.55	2.98	T
2023-Q3	478.8	464.8	452.8	2.96	5.58	T
2023-Q4	482.3	457.4	449.6	5.30	7.02	T
2024-Q1	506.8	485.2	470.9	4.35	7.33	T
2024-Q2	504.9	470.5	463.1	7.07	8.63	T

Backtest detail, Expedite cost proxy (PPI: Couriers & Express, FRED) (continued)

Quarter	Actual	Baseline	Δ.72	sAPE base %	sAPE Δ.72 %	Gate
2024-Q3	504.7	478.8	471.1	5.28	6.89	T
2024-Q4	510.1	482.3	475.5	5.60	7.02	T
2025-Q1	540.7	506.8	494.5	6.48	8.93	T
2025-Q2	537.5	504.9	495.9	6.25	8.05	T
2025-Q3	545.7	504.7	498.1	7.80	9.12	T

Appendix, CSV exports

- open_ops_proxies_quarterly_raw_and_delta72_2020_2025q3.csv (quarterly series raw + Δ.72)
- delta72_summary_block_open_ops_proxies.csv (Δ.72 summary blocks)
- backtest_detail_TSIFRGHT_q_2020_2025q3.csv (backtest detail)
- backtest_detail_CASS_SHIP_q_2020_2025q3.csv (backtest detail)
- backtest_detail_PPI_DELIV_WHSE_q_2020_2025q3.csv (backtest detail)
- backtest_detail_MFG_OT_HRS_q_2020_2025q3.csv (backtest detail)
- backtest_detail_DELIVERY_TIME_DIFF_q_2020_2025q3.csv (backtest detail)
- backtest_detail_PPI_EXPEDITE_q_2020_2025q3.csv (backtest detail)

Sources: FRED (Federal Reserve Bank of St. Louis) series TSIFRGHT, FRGSHPUSM649NCIS, CEU3000000004, PCUADLVWRADLVWR, PCU4921104921102, DTCDFNA066MNFRBPHI.